

Chapter 13

Ultratrace Impurity Analysis of Wafer Surfaces

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13.1 INTRODUCTION

13.1.1 Analytical Measurement of Silicon Surfaces

Production of semiconductor devices involves exposing wafers to a wide range of environmental conditions during a multitude of production steps. Each of these steps may deposit unintentional contamination upon on the wafer surface, detrimentally impacting device performance. Since revenue only is generated from devices operating within relatively narrow performance windows, any impact of contamination on performance must be minimized. With shrinking device dimensions, contamination-monitoring programs are becoming an increasingly critical requirement of any Si processing facility.

Effective contamination monitoring programs have several metrology requirements that must be met in order to adequately support state-of-the-art device fabrication. The metrology must be able to quantify or qualitatively identify the contamination, that is, to determine the chemical identity of the contaminant. It must be able to quantitate the contaminant, either in an absolute (the exact number of atoms or molecules present) or in a relative sense (relative to an accepted “standard” value). For the majority of process control applications, a relative measurement is completely adequate and is much easier to obtain. Absolute quantitation is generally required only when compared to other facilities employing different metrology techniques or to contamination level guidelines, such as those defined in the International Technology Roadmap for Semiconductors [1]. The metrology used for a particular analysis must be sensitive enough to detect contamination below the levels that impact device performance.

13.1.2 Common Properties of Contamination in the Semiconductor Environment

Contamination of interest to Si processing may be divided into several broad categories. The broadest classifications are inorganic and organic contaminants. Measurement of inorganic contamination is further subdivided. Elemental analysis, by far the most common, is when the concentration of an element is measured, regardless of its chemical state; e.g., the concentration of Fe is quantified, in units of atoms/cm², whether it is present as Fe⁰, FeO, Fe₂O₃, or Fe(OH)₂. The second broad classification of inorganic analysis is the

measurement of distinct chemical compounds. An example is the measurement of nitride as ammonia, NH_3 , distinct from other N-containing species, such as Si_3N_4 .

Because organic compounds possess a limited diversity of atoms but a wide diversity of chemicals, metrology is focused on specific chemical compounds or a limited class of compounds (such as amines or alkyl hydrocarbons). Exceptions to this might include the measurement of total organic carbon on pre-gate oxide wafers, where a broad range of organic compounds may pyrolyze in the gate furnace and react with the Si substrate to form detrimental SiC islands [2]. Additionally, metrology may determine if the contamination is present as a relatively homogenous film or as discrete islands or particles; such information may be useful in identifying and eliminating the source of the contamination.

Contamination may be transferred from various sources to a wafer through a wide range of modes as discussed in Chapter 2. Any variation in contamination levels must be detected and examined, because even an inadvertent reduction in existing levels may not necessarily be beneficial. An example of this would be boron contamination from degradation of borosilicate particle filters. This may result in a background doping level in the device active region that was compensated for by counter-doping during production recipe optimization. If the B-doping levels were substantially reduced, for example by replacement of the filter with fluoropolymer-based media, the counter-doping would now be excessive and result in yield loss. Therefore, it is imperative that the baseline contamination levels for all species of concern be known and constantly monitored.

13.1.3 General Selection Criteria for Ultratrace Metrology

The selection of the analytical technique to apply to a given situation is greatly dependent upon the information required. A valuable approach to take is to define the question that needs to be answered, if that question to be considered is “What is the level of elemental contamination on daily monitor wafers processed through a set of several nominally clean diffusion furnaces?” The appropriate metrology would be vapor phase decomposition/inductively coupled plasma–mass spectroscopy (VPD/ICP–MS) with its ability to quickly analyze the entire wafer surface with extremely low detection limits for a wide range of elements. If the need were to examine the edge (within 200 μm) of a wafer surface with 10^9 atoms/ cm^2 detection limits for Li, Al, P, and Cr, then time of flight–secondary ion mass spectroscopy (TOF–SIMS), with its high lateral resolution and low detection limits for the majority of elements in the periodic table, would be the most suitable choice. Elemental identification of 0.5- μm diameter particles distributed over a wafer would be the forte of Auger electron spectroscopy (AES) with its ability to perform survey image scans using secondary electron emission followed by ultrahigh lateral resolution elemental analysis.

Some general guidelines on the selection of which technique to use are given below:

- For fast and simple elemental analysis of transition elements with detection limits on the order of 1010 atoms/cm², perform a 3–5-spot analysis using total reflection X-ray fluorescence (TXRF). A caveat is that inhomogeneous contamination may be missed when it does not lie within one of those analyzed areas, as shown in Fig. 13.1-1. If full wafer analysis is needed for these elements, use VPD sample preparation with subsequent TXRF of the resulting spot. The detection limits will also be lower, due to the concentrating effect of the droplet scanning.
- If full wafer analysis with 108 atoms/cm² detection limits is needed, perform VPD/ICP–MS.
- If elemental particle analysis is required, perform AES or TOF–SIMS. Energy dispersive X-ray analysis (EDS) may be used, but be aware that for small particles (<5 μm diameter), the majority of the analytical signal is coming from the region outside of the particle itself.
- If chemical state information is needed, e.g., to determine if the contamination is Fe, FeO, Fe₂O₃, or Fe(OH)₂, X-ray photoelectron spectroscopy (XPS) is the technique of choice. It has lower spatial resolution and is not as rapid as AES, but provides chemical state information and is less destructive than AES when analyzing polymers.
- If there is organic contamination, TOF–SIMS is the technique of choice. Attenuated total reflectance–Fourier-transform infrared spectroscopy

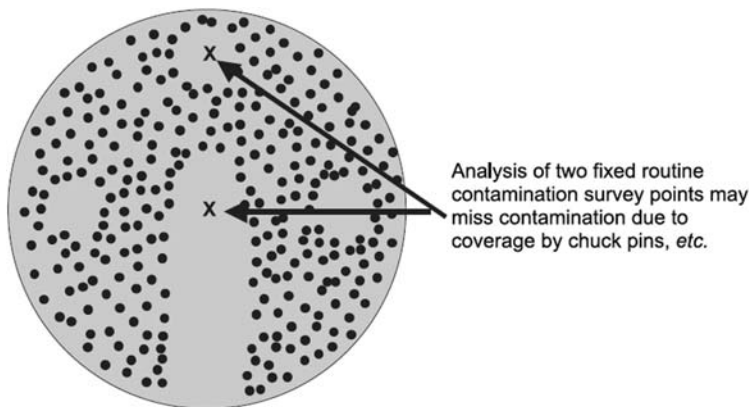


FIGURE 13.1-1 Spot analyses techniques may miss inhomogeneous contamination. *Used with permission from the authors.*

(ATR—FTIR) may also be helpful if a sufficient amount of the material is present.

- To determine the nature and amount of volatile contamination deposited on the entire wafer surface from the ambient atmosphere, use thermal desorption followed by gas chromatography—mass spectrometry analysis of the desorbed species.

There are several stages in the selection and performance of wafer surface analysis:

1. *Pre-planning*: Planning should be based on discussions with the requestor as to the scope of their analytical needs and the detection limits needed to address their contamination issue. Purity of required chemicals and materials required for the analysis must also be considered at this stage. Availability of certified standard materials, if required, must be addressed.
2. *Sample Collection and Storage*: Appropriate sampling times and methods of collection must be determined. Collection protocols and sample storage issues must be considered.
3. *Sample Preparation*: This stage must be considered carefully with respect to the purity levels and cleanliness required to address the customer's needs. Chemical purity and preparation techniques must be adequate to ensure no contamination is added during sample preparation.
4. *Sample Measurement*: Stable, reliable, and accurate methods must be employed. Any spectral or other matrix interferences must be determined and remediated.
5. *Calculation and Reporting of Data*: The error limitations of the chosen analytical technique need to be considered, relative to the customer's error tolerances.

13.1.4 Advantages and Limitations of Surface Contamination Measurement Techniques

The limitations inherent in ultratrace metrology techniques must be considered when selecting an appropriate analytical technique and may be segregated into two broad categories: sensitivity and spatial resolution. The sensitivity of the technique to the analyte of interest determines if the technique can detect the particular species and, if so, whether it can be detected with a sufficient level of sensitivity. Spatial resolution requirements determine whether the technique can measure the analyte with the required lateral and depth resolution. When analyzing a single 0.1- μm diameter particle using a whole wafer method, such as VPD, the analysis may not assure that any detected contamination arises solely from that particle, but can be caused by contamination elsewhere on the wafer. [Table 13.1-1](#) summarizes the key advantages and limitations of each analytical technique.

13.2 INORGANIC CONTAMINATION ANALYSIS TECHNIQUES

13.2.1 Vapor Phase Decomposition

Vapor phase decomposition is a technique that provides high sensitivity, full wafer analysis. The VPD process is accomplished by exposing an unpatterned silicon wafer (with a native or other oxide layer on the surface) to HF vapor in a closed, chemically inert container, resulting in the decomposition of the oxide into HF-containing H₂O microdroplets across the wafer surface that trap the elemental contaminants present on the surface. A small droplet of ultrahigh-purity scanning collection solution (SCS) is then translated across the wafer surface to concentrate all the microscopic droplets, creating during fuming, into a single sample with a volume ranging from 100 to 1000 μ L. Once prepared, the liquid sample is analyzed by instrumental techniques for qualitative identification and quantitative analysis of contaminant originally present on the wafer surface. Sample preparation by VPD can be coupled with several different trace elemental analysis techniques, such as atomic absorption spectroscopy (VPD/AAS), total reflection X-ray fluorescence (VPD/TXRF), and inductively coupled plasma mass spectrometry (VPD/ICP–MS) [3]. Manual collection of the droplets from the wafer's hydrophobic surface is one of the problems with traditional VPD sample preparation, but advances in robotics and automation have allowed several analytical equipment suppliers to create fully automated and semiautomated VPD systems [4]. For manual collection, the wafer surface is decomposed and a vacuum pencil holds the wafer while wrist motions scan the SCS droplet across the wafer surface. The objective of the manual scanning process is a uniform, systematic movement of the SCS droplet across the entire wafer surface. A high level of skill and experience is required to achieve this goal with consistency. Inherent problems include sample preparation precision, total surface coverage, wafer-edge contamination, and sample loss during the scanning process. With the advent of 300 mm wafers, manual scanning of the wafer surface has become prohibitively expensive. Automated robotic scanners are now capable of highly reproducible and controlled movement of the SCS droplet across the entire wafer surface [5]. End-users must balance the degree of automation required with the costs that are incurred. Manual VPD systems are the least expensive, but suffer greatly from reproducibility and throughput issues. Regardless of the scanner's configuration, there must be a mechanism for the quantitative and contamination-free transfer of the SCS droplet into a sample vial for subsequent instrumental analysis.

Recent changes and advancements in VPD equipment includes various equipment suppliers creating a higher degree of automation and coordination with computer integrated manufacturing systems, overhead transportation systems, and metrology equipment allowing for the automatic and simultaneous

TABLE 13.1-1 General Characteristics of Various Analytical Techniques Used to Analyze Contamination on the Surface of Si Wafer				
Technique	Application	Advantages	Limitations	Detection Limit
Total Reflection X-ray Fluorescence (TXRF)	Elemental analysis (Z > Si)	Highly automated, nondestructive, rapid analysis oxides and nitride films analyzed same as bare Si	Higher detection limits than VPD methods cannot analyze Na and Al, limited analysis area	Low 10 ¹⁰ atoms/cm ²
Vapor Phase Decomposition/ Inductively Coupled Plasma Mass Spectroscopy (VPD/TXRF), VPD/ Atomic Absorption Spectroscopy (AAS)	Elemental analysis	Low detection limits, ability to analyze low Z elements	Destructive analysis	Low 10 ⁸ atoms/cm ²
		Analysis of entire wafer surface (partial sections may also be analyzed)	More labor intensive than TXRF	
			Higher analyst skill level required (lowest for VPD/ TXRF). Scanning droplet chemistry may impact collection efficiency	
Time-of-Flight Secondary Ion Mass Spectrometry (TOF—SIMS)	Organic, complex ion, and elemental analysis	Very high lateral (1000 Å) and depth resolution (<10 Å)	Small analysis areas requires a high-analysis spot density to give results representative of surface	Low 10 ⁷ –10 ⁹ atoms/cm ² for elemental species

Continued

TABLE 13.1-1 General Characteristics of Various Analytical Techniques Used to Analyze Contamination on the Surface of Si Wafer—cont'd

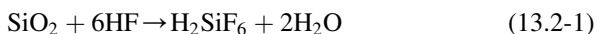
Technique	Application	Advantages	Limitations	Detection Limit
		Low detection limits	Quantification may be nontrivial	
		Measurement of multiple species in a single analysis may be used in survey, imaging, and depth profiling modes		
Fourier Transform Infrared Spectrometry (FTIR), Attenuated Total Reflection FTIR (ATR–FTIR)	Analysis of FTIR active species, typically organic species	Provides information on chemical nature of organic species	Relatively high detection limits compared to TOF–SIMS or thermal desorption Well suited for large (>10 μm) particle analysis	0.1–100 ppm
Auger Electron Spectroscopy (AES)	Elemental analysis ($Z > \text{Li}$)	Very high lateral resolution and surface specificity may be used in survey, imaging, and depth profiling modes, excellent for particle analysis	High detection limits makes AES unsuitable for ultratrace analysis	0.1–1 at. %

Energy Dispersive X-ray Analysis (EDS)	Elemental analysis	Rapid analysis easily combined with electron microscopy	Large analytical volume allows easy data misinterpretation	0.1–1 at.%
			EDS is unsuitable for ultratrace analysis due to high detection limits	
X-ray Photoelectron Spectroscopy (XPS)	Chemical state analysis of elemental species	Provides chemical state information on contamination, relatively nondestructive on polymer particles	Significantly higher analysis times than AES XPS unsuitable for ultratrace analysis due to high detection limits. Lateral resolution not as low as AES	0.1–1 at.%
Thermal Desorption Analysis Gas Chromatography Mass Spectrometry (TDA–MS)	Analysis of adsorbed organic species	Analysis of entire wafer surface	Provides no information on areal distribution of contamination	5×10^{11} C atoms/cm ²

operation of the VPD and an ICP—MS. This improvement also helps to reduce adventitious contamination and increase the throughput of the equipment as operators do not manually transport samples between the VPD and the ICP—MS or take valuable time and resources to load wafers onto the VPD equipment [5].

13.2.1.1 Hydrofluoric Acid Decomposition Stage

During the VPD process, the surface oxide layer on a wafer (native, chemical, or deposited, typically less than 200 Å thick, although thicker films may be analyzed under more aggressive conditions) is decomposed by exposure to HF gas according to the following equation [6]:



Vapor HF is produced on-demand by bubbling ultrapure N₂ or another inert gas through a liquid reservoir containing concentrated aqueous HF solution. The inert gas displaces HF molecules in solution releasing them into the flow stream. The decomposition rate of the SiO₂ is dependent on the HF concentration of the solution and the resulting gas phase concentration of the HF inside the fuming apparatus [6]. This reaction of HF with SiO₂ on the surface of the wafer releases impurities present on the surface into a H₂O layer formed by decomposition of the oxide.

Some manufacturers use various configurations of a Peltier device or chilled liquid flow to cool the wafer [7], thus allowing better condensation and collection of the gaseous HF on the SiO₂ surface. Finally, the type of oxide plays an important role in the decomposition rate; native oxides and steam oxides decompose faster than thermal or high-temperature annealed SiO₂. Etching or decomposition occurs only if a condensed HF-containing liquid layer is present on the surface of the wafer. Therefore, an important part of the VPD process is optimizing conditions of gaseous HF concentration, temperature, and H₂O partial pressures that lead to the largest amount of condensation on the surface of the wafers. Initially, an SiO₂ surface is exposed to a vapor phase mixture of H₂O and HF. If their partial pressures are sufficiently high, then a condensed film of HF and H₂O forms and continue to accumulate on the SiO₂ surface. Etching proceeds with the formation of H₂SiF₆ and additional H₂O. Although the partial pressure of SiF₄ over dilute H₂SiF₆/HF/H₂O is relatively low, some SiF₄ does evolve during etching. When the etching is complete and the HF flow halted, the liquid H₂SiF₆/HF/H₂O film can be evaporated or collected off the surface of the wafer [8] for subsequent analysis.

13.2.1.2 Wafer Scanning Process

Once the surface oxide has been decomposed, a scanning technique is employed wherein a solution of HF and H₂O₂ or HNO₃ in ultrapure H₂O (UPW) is pipetted onto the center of the wafer and the droplet is translated across the wafer surface to collect and concentrate the impurities liberated

during the decomposition. For maximum extraction efficiency, the scan may be performed multiple times, from the center to the edge and a return pass from the edge to the center. When using manual scanning, the analysts must be extremely precise in their manipulation of the SCS droplet across the surface of the wafer using only a wafer wand and wrist movements. When using automated systems, if a wafer scribe is present on the wafer surface, an edge exclusion is typically employed to prevent the droplet from being trapped by the scribe. Once the droplet has returned to the center, it is collected and transferred to an ultraclean sample vial, which may be capped to prevent volume loss by evaporation. The VPD instrument must be located in a cleanroom area where the background contamination of the droplets by the laboratory ambient atmosphere is lower than the required detection limits for the technique. Additionally, full cleanroom gowning and protocols are recommended to prevent adventitious contamination from collecting on the wafer surfaces.

13.2.1.3 Collection Efficiency of the Scanning Process

After VPD, the elemental contamination can be solvated in the VPD solution or redeposited onto the wafer surface by one of several mechanisms. First, the metal ions can physically adsorb to the wafer surface following Langmuir equilibrium [7]. This process is not preferred due to the limited number of negatively charged sites present on the hydrogenated wafer surface after the VPD process. The precipitation of metal ions through condensation with SiF_6 is a possible concern, but has not been demonstrated [8]. Another issue arises with metals more noble than Si, such as Cu, Ag, Au, and Pt; these elements can easily be electrolytically reduced to their metallic form by oxidizing the denuded Si. Therefore, the collection efficiency of the VPD process depends on the ability of the scanning droplet to remove all the contamination by trapping the small droplets and by dissolving all the readsorbed contamination from the wafer surface.

Ideally, the collection solution should be sufficiently acidic, and thus complexing, to maintain the metallic cations in their ionic form. The oxidizing potential of the scanning solution should be high to oxidize the noble metals previously mentioned. A variety of collection solutions may be used for this purpose. The most common are UPW, diluted HF, or diluted HNO_3 , with or without H_2O_2 . The use of scanning solutions containing HNO_3 can add N-mass interference to any quadrupole-based ICP–MS systems; thus, application of dilute HF with H_2O_2 instead of HNO_3 as the oxidant is preferred. Recoveries are generally the lowest with UPW alone, which is acidified by the residual HF remaining on the wafer. Nitric acid is a strong acid and a good oxidizing agent, but as noted, mass spectroscopy results suffer from N-based interferences. Hydrofluoric acid is a weak acid, but has good complexing properties. The acidification by low-concentration HF in the UPW provides

little complexing or oxidizing power to remove metals with higher potentials than Si (-0.16 V).

The addition of a strong oxidant such as HNO_3 or H_2O_2 is necessary to oxidize and dissolve metals that are nobler than Si. The kinetics of noble metal dissolution are the result of the competition between their deposition due to Si oxidation and their removal via metal reoxidation. Copper is a good example of a metal whose collection efficiency is dependent on the oxidation power of the VPD scanning solution to keep it in solution. For other metals, high collection efficiency may be obtained with diluted chemistries, which offer good compatibility with the analytical techniques used for the SCS solutions. If the acid concentration of the scanning solutions is above 15 vol% (by volume) then further dilution may be needed to prevent damage to the analytical instrumentation. For high levels of contamination, the collection efficiency may decrease for some metals due to the saturation of the collection solution.

Wafer surfaces may not be completely hydrophilic after the vapor phase etching process due to roughing caused by the VPD process or the underlying substrate or the film having composition that inhibits the formation of a hydrophilic surface. In this case, a process incorporating a scanning nozzle, which uses a vacuum to hold the SCS drop inside the nozzle, may be used to scan the droplet across a nonhydrophilic wafer surface, instead of relying upon surface tension alone to maintain the droplet shape. These dual nozzle systems also allow for the automated scanning of the wafer edge or bevel, as the outer edge of the wafer often contains contamination that can transport or diffuse to devices and, therefore, is of interest [9,10].

13.2.1.4 Possible Contamination Sources During the VPD Process

Ultrapure Water: Contamination in the UPW used to prepare calibration standards and SCS solutions may present a significant contribution to background levels and impact the detection limits of the VPD technique. For 10^8 atoms/cm² detection limits, the UPW must be 18 M Ω or better. Point-of-use H_2O purifying systems are also needed to further purify UPW and facilitate the removal of B and organic contamination. Typical background levels in the UPW must be in the parts-per-trillion-range if low 10^8 atoms/cm² detection limits are to be achieved.

Chemical Vessels: All vessels used for the VPD process must be cleaned in acid solutions. Most polymers must be leached of metallic contaminants prior to use. Quartz or borosilicate glass vessels cannot be used due to the high background contamination. Most vessels should consist of some type of fluoropolymer, or a polymer of ethylene or propylene.

The VPD Chamber: Another critical design feature is the use of chemically inert, high-purity polymer construction materials for the VPD chamber to prevent sample contamination and to withstand the harsh chemical

environments of the VPD process. The chamber must be constructed of highest purity fluoropolymeric material with low metallic contamination. To aid in maintaining an ultraclean environment inside the VPD chamber periodic cleaning and a break-in period with exposure to gaseous HF for long periods (a passivation step) ensures low background levels from the high-purity polymeric materials [11].

Chemical Purity: The purity of all solutions used for scanning, cleaning, and calibration directly impacts the background noise of the measurement and the ultimate detection limits. The chemicals have to be of high and consistent purity. Metallic contamination levels in the acids and peroxide used to prepare the SCS solutions should be in the 10 pg/ μ L range in order to achieve and maintain 10^8 atoms/cm² detection limits.

13.2.1.5 Instrumental Analysis of the VPD Droplet

ICP–MS: For ICP–MS analysis, the SCS solution collected from the VPD process is nebulized into an Ar plasma. The contaminants that are dissolved into the scanning solutions are vaporized, ionized, and extracted into a mass spectrometer for analysis. One of the advantages of ICP–MS is the wide range of elements, from Li to U, which can be analyzed in a single analysis. This sensitive, rapid, multi-element mass spectroscopic technique offers ultralow detection capability, high sample throughput, wide dynamic range, wide elemental coverage, rapid quantitative analysis, and the ability to provide isotopic information. With its relatively simple spectrum, free of organic interferences, the spectrometer is rugged, reliable, and stable and can be used to run a wide variety of sample matrices. One shortcoming of the Ar-plasma ICP–MS is instrument-caused background interferences in the analysis of ⁴⁰Ca, ³⁹K, and ⁵⁶Fe caused by the formation of ⁴⁰Ar⁺, ³⁸Ar¹H⁺, ⁴⁰Ar¹⁶O⁺, and other plasma-derived complex ions that result in elevated background levels. However, it has been shown that the Ar and Ar-polyatomic levels may be significantly reduced by operating under cold plasma conditions [12] (i.e., low ICP power and high nebulizer gas-flow rate) or the use of collision or reaction cell technologies [13]. Cold plasma conditions lower the plasma temperature, which drastically reduces the formation of Ar-based molecular species and allows the determination of elements that otherwise could not be determined at ultratrace levels with ICP–MS. The ICP–MS may be configured to analyze both cold plasma elements (e.g., K, Ca, and Fe) and normal plasma elements (e.g., W, Mo, and Cu) in the same analysis, using a low flow rate nebulizer optimized for low sample volumes.

However, with the very small volumes associated with VPD extraction, there is a fine balance between elemental coverage and detection capability. Additionally, because the available ionization energy is much lower in cold plasma, matrix interferences and polyatomic spectral interferences from the Si matrix of the VPD sample may present a potential problem. For this reason, a

combination of cold and normal plasma, even with a nebulizer flow rate optimized for small samples, may not be the optimal approach for VPD extracts. Limited sampling analysis of such extracts requires minimum matrix/polyatomic interferences that are isobaric with the analytes of interest, sub-parts-per-billion quantification, and expanded elemental coverage. When using an ICP–MS, the analyst must determine the starting sample volume of the scanning droplet, and then choose between the number of elements quantified and the respective detection sensitivities. Due to the low sample volume of the scanning solution (100–1000 μL), low flow rate nebulizers should be used to allow analysis of a wide range of elements with sufficient data acquisition time. The calibration of the ICP–MS should involve the analysis of 3–4 multielement calibration standards in the concentration range of interest. The correlation coefficients of each individual calibration curve, the instrument blank concentration, and the concentration of elements in the quality control (QC) sample are then compared. If all parameters are within the acceptable limits, the analytical sample droplets are then analyzed. Additional instrumental blanks and multielement QC samples are analyzed at the beginning, middle, and end of each analysis set to verify that no calibration anomalies or instrument contamination has occurred.

VPD/AAS: AAS is another elemental analysis technique for analyzing the SCS solution obtained during VPD. It is a simpler and more cost-effective technique than ICP–MS when analyzing a large number of samples for a very small number of elements. However, AAS analysis is applicable for fewer elements than ICP–MS and some of these elements require special preparation to minimize chemical and physical interferences. The theory behind AAS is that elements in their elemental form absorb ultraviolet light (UV) when they are excited by heat. Each metal has a characteristic absorption wavelength that may be used to identify it. The AAS instrument targets a particular element by focusing a beam of UV at a specific wavelength through the sample. The sample is volatilized in a flame or a heated graphite tube and then flows into a detector. If a metal is present in the sample, it will absorb some of the light and thus reduce the intensity of the UV light; this change is first quantified and then converted into a concentration by comparison with known standard solutions. Some of the difficulties in the variation of volatilization conditions may be circumvented by inclusion of a known concentration of a similarly emitting element and relating the concentration ratios of the emission intensity of the standard to the unknown.

VPD/TXRF or TOF–SIMS: The use of VPD as a preconcentration step with subsequent analysis of the droplet area with TXRF or TOF–SIMS [14,15] has proven to be a useful method to provide detection limits of these techniques to low levels (e.g., 10^8 atoms/ cm^2) for Fe using TOF–SIMS analysis [14].

13.2.2 Total Reflection X-Ray Fluorescence Spectrometry

X-ray fluorescence [16] is a phenomenon in which a primary X-ray photon of sufficient energy is absorbed by an atom, which reacts by ejecting a core-level electron, as shown in Fig. 13.2-2. This is termed the “photoelectric effect” and the ejected electron is called a “photoelectron.” The vacancy created by the ejection of the photoelectron leaves the atom in an energetically unstable state. Atoms prefer to have as low an energy as possible and they achieve this when a higher energy level electron falls down to fill the lower energy position previously occupied by the ejected photoelectron. When this occurs, the atom has an excess of energy equal to the difference of the energy level previously occupied by the electron and the lower level, originally occupied by the photoelectron. This excess energy is relieved in one of two ways: (1) by the ejection of a second outer shell electron (the Auger electron), where the kinetic energy of the Auger electron is the energy difference of the two levels or (2) by the production of an X-ray fluorescence photon, whose energy ($h\nu$) is the energy difference. These two effects are competitive, with AES more probable for lower atomic number elements and XRF for higher atomic number elements [17], as shown in Fig. 13.2-3.

Each element has distinct energy levels for its electrons; by measuring the energy of the Auger electron or X-ray photon, the atom that give rise to the particular energy is identified. Because there are a variety of levels in the atom that are involved in these transitions, each element can produce not one, but a series of Auger electrons or X-ray photon energies that are characteristic of that element. In order for the core vacancy to be generated, the incident primary X-ray photon must have a higher energy than the binding energy of the photoelectron; this energy is commonly termed the absorption edge and is unique for that element. For TXRF analysis, a primary X-ray source that produces an X-ray photon energy above the absorption edge for that element

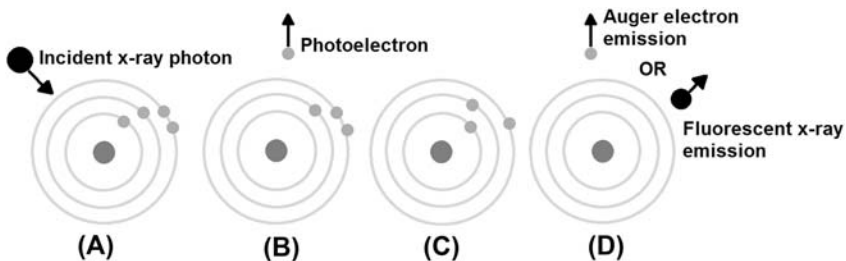


FIGURE 13.2-2 Schematic of X-ray fluorescence process: (A) an incident primary X-ray photon is absorbed by the target atom, (B) the atom responds by ejecting a core electron termed the “photoelectron” leaving a vacancy in the lower core level, (C) an electron from a higher energy level falls to fill the created vacancy leaving the atom in an unstable state, (D) the atom relaxes by emitting the excess energy as the kinetic energy of an ejected Auger electron or the energy of a fluorescent X-ray photon. *Used with permission from the authors.*

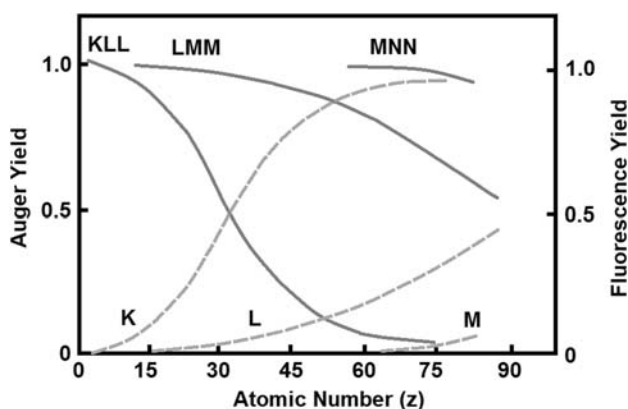


FIGURE 13.2-3 Auger and X-ray fluorescence yields as a function of atomic number [17]. Used with permission from *Handbook of Auger Electron Spectroscopy, Third Edition*.

must be used. It may be argued that it would be wise to simply use a very high-energy X-ray source to encompass all elements being analyzed. The concern is that the fraction of the primary X-rays absorbed by the wafer, and hence ultimately converted into fluorescent photons, is inversely proportional to the photon energy; the higher the energy, the lower the amount absorbed. Therefore, to maximize the sensitivity, the optimum process must exceed the highest absorption edge expected to be encountered, but to exceed it by as little as possible. In order to accommodate a variety of conditions, most modern TXRF instruments allow for a variety of different source targets to be used for analysis. Tungsten is a common target material because its L series fluorescence lines lay in the 10-keV range directly above the K series lines for transition elements. Additionally, the primary X-rays are generated by bombarding the target metal with a high-energy electron beam, which impacts a great deal of energy to the target, significantly heating it. Since W has a very high melting point, it may be subjected to a higher current electron beam than many other metals without melting. The higher electron current generates a higher primary X-ray flux, more rapid analysis times, and lower detection limits.

If the primary X-ray beam strikes the sample above the critical angle (the angle below which total reflection occurs) the X-rays penetrate through the Si wafer, giving rise to a significant background signal due to the fluorescence of the Si and Bragg scattering of the primary X-rays from the Si crystal lattice, as shown in Fig. 13.2-4. If the beam is directed to impact the sample below the critical angle, total reflection [18] occurs and the X-rays do not penetrate into the sample more than a few tens of angstroms, resulting in a significantly reduced background signal.

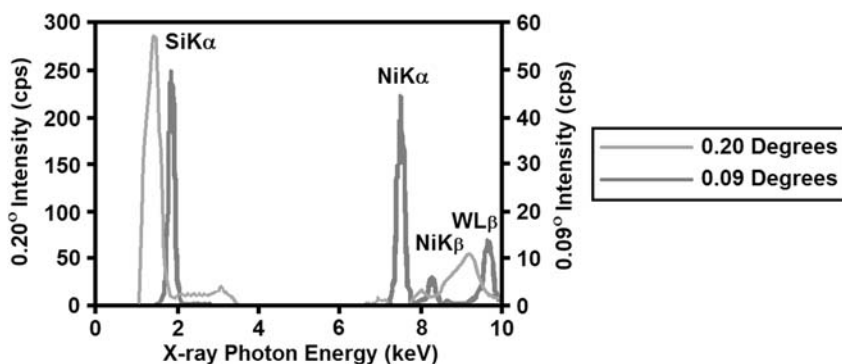


FIGURE 13.2-4 Impact of primary X-ray incidence angle on TXRF analysis of an Si wafer having 10^{13} atoms/cm² Ni contamination; note the difference in intensities between the two scans. *Used with permission from the authors.*

This reduction of the background signal allows the fluorescent peaks of contaminant atoms to be resolved at much lower concentrations than would otherwise be possible. One significant advantage of TXRF is that the components of the two most common Si-containing films in semiconductor manufacturing, O and N (in SiO₂ and Si₃N₄), are of very low atomic number and, consequently, are essentially invisible to the XRF process. This allows SiO₂ and Si₃N₄ films of any thickness to be analyzed in exactly the same manner as bulk Si. Wafers containing blanket metal films or metal patterned wafers are not suitable for analysis by TXRF, due to the high peak signal levels and the resulting dead time (the recovery time between photon arrival events over which the detector is unable to detect the arrival of a photon) of the detector. A typical TXRF instrument is shown in Fig. 13.2-5.

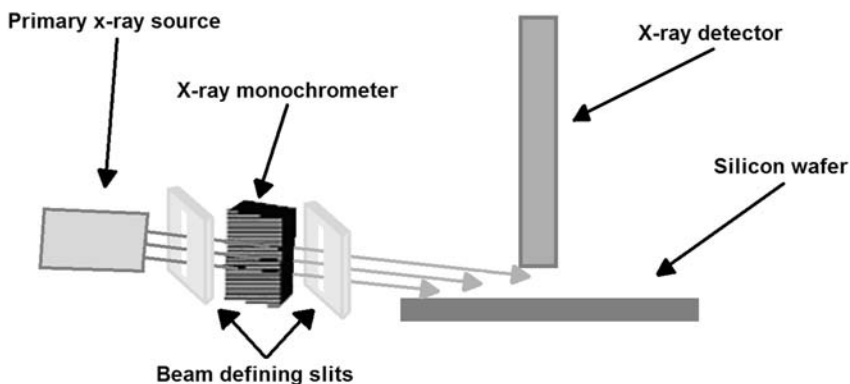


FIGURE 13.2-5 Schematic diagram of TXRF instrumentation. *Used with permission from the authors.*

The source of primary radiation may either be a simple sealed tube, a rotating anode, or a synchrotron. In a sealed tube source, a beam of energetic (tens of keV) electrons generated from a heated W wire filament is directed onto a H₂O-cooled metal target. These electrons strike the metal atoms composing the target, creating core holes and generating secondary X-rays at the characteristic energy of the metal. The higher the electron current striking the target, the higher the primary X-ray flux and higher the sensitivity, limited by the melting point of the metal. A commonly used method to achieve higher incident electron currents without melting is rotating the anode. For this type of source, the metal is in the shape of a cup with the electron beam striking only a small area at any one time as it rotates. Once a given area of the target has been impacted, and heated, it has the full rotation of the cup to cool before being heated again [16].

A synchrotron is a highly specialized circular particle accelerator that produces an extremely bright X-ray beam of tunable energy. The use of synchrotron radiation (SR) greatly extends the utility of TXRF for Si wafer contamination analysis [19]; however, due to the limited availability, large size, and expense, it is not suitable for high volume routine contamination monitoring. The advantage of SR versus X-ray emitted from tubes, rotating anodes, and liquid metal X-ray sources is the continuous spectrum and intensity that is several orders of magnitude higher compared to the highest powered X-ray tubes. SR–TXRF can therefore be used to successfully detect very small numbers of atoms (1×10^7 atoms/cm²) in very small spatial areas [20]. The higher flux and brightness also allows for chemical state information of the metallic impurities to be determined using X-ray adsorption near edge structure [21].

A recent approach to improve the sensitivity of TXRF and to providing a higher flux primary X-ray illumination, without using SR sources, has involved the application of liquid metal X-ray sources in which the target material is a flowing stream of self-healing liquid metal, such as a gallium-indium alloy, which is liquid at room temperature and allows operation at very high power levels [22,23] up to 100× of conventional X-ray sources [24] with a commensurate increase in sensitivity.

However generated, the primary X-ray beam is defined by one or more slits, which act to define the shape of the beam by blocking the X-ray photons, except those that can pass through the slit, and then are passed through a crystal monochromator to remove extraneous X-rays, except those of the desired target fluorescence energy, which significantly reduces the background signal resulting from scattering of Bremsstrahlung radiation. The latter is a continuum X-ray background generated by the X-ray source due to the abrupt kinetic energy loss of the electron decelerating in the metal target, reflecting off the sample, and entering the detector. The primary X-rays then strike the Si wafer below the critical angle, exciting contaminant atoms on its surface to produce characteristic X-rays fluorescence. Detection of the fluorescent X-rays

is accomplished with a lithium-drifted silicon (SiLi) detector [25], where the incoming photons strike a Si crystal that has been doped with Li ions. The interaction generates electron-hole pairs, whose population is proportional to the energy of the incoming photon. The intensity of the photons and the frequency at which the electron-hole pairs combine is maintained such that only one photon enters the crystal during each counting cycle. If two photons enter during a given cycle this is termed a “coincidence event,” and the energy is measured at twice the incoming photon energy and gives rise to an artifact peak in the TXRF spectrum, referred to as a “sum peak.”

TXRF is a very rapid technique, well suited for high throughput analyses, where detection limits below 10^{10} atoms/cm² or of low atomic number elements (<Si) is not essential. With modern instrumentation, the operator simply places a carrier of wafers into the instrument, commences the analysis, and returns to find a stack of report sheets or a computer file listing the results of each analysis area (typically 3–5 per wafer for routine survey analyses). Quantification is performed by comparison to standard wafers of known surface concentration. These standards are typically prepared by spin casting aqueous solutions containing the analytes of interest [26] and concentration verification of like-prepared wafers using a technique, such as VPD/ICP-MS, which may be referenced back to primary standard reference materials.

13.2.3 VPD/TXRF

TXRF analysis alone, without coupling to VPD, is a surface-sensitive technique capable of detecting medium and high-Z elements (sulfur to uranium) at very low levels; detection limits generally are in the 10^{10} atoms/cm² range using a tungsten rotating anode X-ray source. More detail on the TXRF technique is provided in Section 13.2.2. Although this technique alone has proven successful in the determination of surface metallic contamination on the wafer surface, the combination of VPD with TXRF has been shown to improve the detection limit of TXRF by preconcentration of the surface contaminants using VPD. The detection limit improvement can be estimated from the ratio of the total wafer area to the analysis area; improvement of the detection limit of two orders of magnitude may be achieved for a 200-mm wafer. There are some disadvantages in using VPD with TXRF. VPD alone is a destructive technique. TXRF without VPD preconcentration is a nondestructive technique where the wafers are exposed only to soft X-rays that do not disturb the wafer surface, whereas VPD preconcentration prior to TXRF completely decomposes the surface oxide and concentrates the contaminants into one small area. Another disadvantage of using VPD prior to TXRF is that the VPD may add contamination to the wafer and thus produce data that are not representative of the actual contamination. Further, using VPD assumes the contamination on the wafer surface is distributed homogeneously on the wafer.

When performing TXRF analysis, the chemical and physical state of the wafer surface impacts the intensity of the X-rays fluoresced; wafer roughness and the physical structural of the metallic contamination on the wafer surface are the main factors contributing to the intensity of the fluorescent X-rays. A classic technique used in TXRF analysis is termed a rocking scan, which is obtained by measuring the fluorescent X-ray intensity as a function of primary X-ray incident angle. This measurement can elucidate whether the contamination on the wafer surface is bulk, films, or particles. With bulk contamination, the signal intensity increases constantly with the sharpest increases at angles greater than the angle needed for total reflection of the primary X-ray off the surface of the wafer. With a contaminating film, the signal intensity increases as the angle is increased but finds a sharp drop in signal intensity at angles higher than the angle needed for total reflection of the primary X-ray off the surface. Finally, particulate contamination exhibits the same behavior as uniform film contamination, except it shows higher intensity at the reflection angle and continues to fall after the reflection angle is passed. When a wafer is prepared by the VPD technique and the sample droplet is dried on the wafer surface, TXRF analysis is affected by the physical state of the dried droplet on the wafer surface and the glancing angle of the incident X-rays. Experiments [27] have shown that the dried SCS residue is indeed particlelike, thus lower glancing-incident X-ray angles can be utilized to produce higher intensity fluorescence from the contaminants on the wafer surface.

13.2.4 Secondary Ion Mass Spectrometry

SIMS (for an excellent review of this useful technique, please see Ref. [28]) is a surface analytical technique in which a sample is placed within an ultrahigh vacuum chamber and bombarded with energetic primary ions; typically 0.5–25 keV. These primary ions strike and penetrate the surface setting the atoms into motion, similar to the cue ball striking the racked balls on a billiard table, as shown in Fig. 13.2-6.

Some of the atoms set into motion may strike nearby particles located in the outermost 1–2 monolayers, ejecting them off the surface and into the vacuum. A portion of these atoms or molecules have a net electrical charge, either positive or negative, hence the term “secondary ion.” The secondary ions are then accelerated in an electrostatic field to several keV and separated on the basis of their mass. SIMS analyses are divided in two modes: dynamic and static. Dynamic SIMS involves bombarding the surface continuously with an ion beam and monitoring elemental concentrations as a function of depth. SIMS depth profiling has become a well-established technique within the semiconductor industry for characterizing dopant (e.g., B, P, and As) profiles and for calibrating ion implanters [29]. It is the method of choice for contamination analyses where high sensitivity, coupled with high lateral or depth resolution, is required.

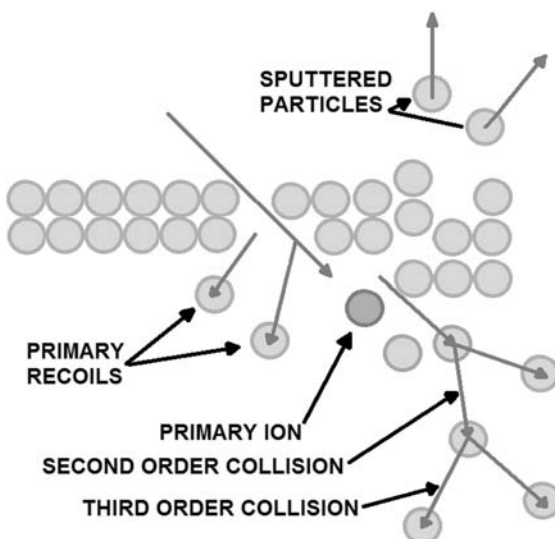


FIGURE 13.2-6 Schematic diagram of the collision cascade initiated by the primary ion impact in secondary ion mass spectrometry. *Used with permission from the authors.*

Since the secondary particles of concern are ions, by definition having a net electric charge, the collection optics and mass analyzers use electrostatic and magnetic fields to manipulate the charge ions. Neutral particles, having no net charge, are unaffected by these fields and do not reach the detector. The fraction of the total number of sputtered particles that are ionized, termed the ion yield, is an extremely critical parameter in determining the actual concentration of an impurity on the sample surface. The major issue concerning the application of SIMS lies with this parameter [30]. The ion yield for a given species may vary several orders of magnitude depending upon the chemical makeup of the surface from which it is sputtered. Because of this extreme variation in ion intensity from the different surface compositions or sputtering matrix, the term “matrix effect” [31] is used. Great care must be taken when determining analyte concentrations from measured ion intensities.

Typically, the analyte signal is measured relative to a presumably unchanging matrix signal, e.g., $^{28}\text{Si}^-$ or $^{30}\text{Si}^+$, when analyzing a Si wafer. An intensity to concentration conversion factor, termed a “relative sensitivity factor” (RSF) [32] for the analyte is applied. By determining the concentration relative to an unchanging matrix species many potential confounding variables, such as changes in primary ion intensities or instrument drift, are removed from the calculation. RSF values must be used with care as they are specific for each analyte and at higher concentrations (above ~ 1 atomic %) may be affected by the concentration of the analyte itself. This matrix effect may be exploited to the analyst’s advantage by providing an environment that

intentionally promotes very high ion yield and, therefore, high analytical sensitivity. Oxide surfaces typically provide a significantly higher positive ion yield for all secondary ions [32], while cesium enhances negative ion yields. For this reason, O and Cs are used as primary ions when analyzing positive and negative secondary ions, respectively. Another variation is to increase the partial pressure of O₂ [32] in the analytical chamber by directing a jet of O₂ gas onto the sample surface during analysis.

Static SIMS [33] involves bombarding a surface with a lower dose, where a primary ion impact has not previously occurred. In the static mode, each primary ion impact produces secondary ions that are characteristic of the original unbombarded surface. Molecular information of organic molecules is preserved when using static SIMS, whereas when using dynamic SIMS the molecules are damaged by the continuous ion beam to the extent that molecular information is lost.

SIMS instrumentation may be divided into two key components: (1) those concerned with the production of primary ions (the primary source and optical column) and (2) those concerned with the collection and distribution of the sputtered secondary ions (the secondary ion column and associated detector). Primary ions are generated by ion sources; these are typical plasma discharge for gas phase materials, such as O₂ or Ar, and thermionic emitters for Cs. When very high spatial resolution is required, field emitter tips using a liquid metal, such as Ga or In, can be used to provide a very bright ion source that can be focused to diameters of 800–1000 Å [34]. The remainder of the primary column involves a series of electrostatic lens, deflectors, and apertures that serve to define the primary ion beam and direct it onto the sample in the desired manner.

Once the primary ions impact the surface, the secondary ions generated by the collision cascade are collected, separated by their mass, and detected by the secondary ion optical column. There are several ways in which this may be accomplished. Magnetic sector instruments [35] separate the secondary ions on the basis of mass by directing the ions through a magnetic field, which serves as a momentum filter (Fig. 13.2-7). Only ions with a specific mass/charge (m/z) ratio have a trajectory, which allows them to pass through the exit slit of the spectrometer and strike the detector. Occasionally, the analyte of concern has an isobaric (same mass) interference, e.g., ³⁰Si¹H[−] and ³¹P[−] at 31 atomic mass units (amu), which presents a background signal that could limit the detection limit of the analyte. When this occurs, the interference may be separated from the analyte on the basis of their respective mass defects. In order to do this the mass spectrometer must have a very high mass resolution (the ability to separate two ions of similar mass). To achieve this high resolution, the ions entering the magnetic sector must have a very narrow distribution of energy; this is accomplished by having both an energy and a mass filter in the beam line, hence, the term “double focusing.” Both energy and momentum (mass) are focused to ensure that the ions of the same mass have

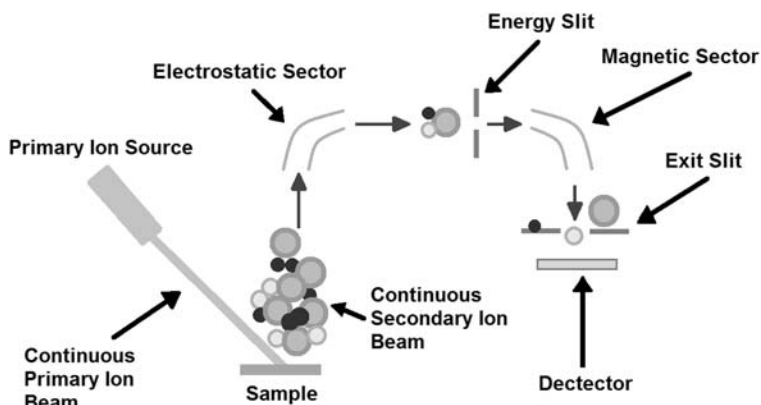


FIGURE 13.2-7 Schematic representation of a double-focusing SIMS instrument. *Used with permission from the authors.*

the same trajectory in the magnetic sector. The exit slit is then made correspondingly narrow in order to allow ions that pass through that small trajectory, and hence of the same mass, to be detected.

A second mass-filtering device for secondary ions is the quadrupole mass filter [28] (Fig. 13.2-8). In this device the secondary ions are passed into a region surrounded by four metal rods, or “poles.” A direct current (DC) voltage is applied to two of these rods and a radio frequency voltage is applied to the other two. The secondary ions are sent into oscillation by the interaction of the DC and RF (radio frequency) voltages. By varying either the DC voltage or RF frequency, ions of different masses may be selected for analysis, with only one mass/charge ratio having a stable trajectory that allows it to pass

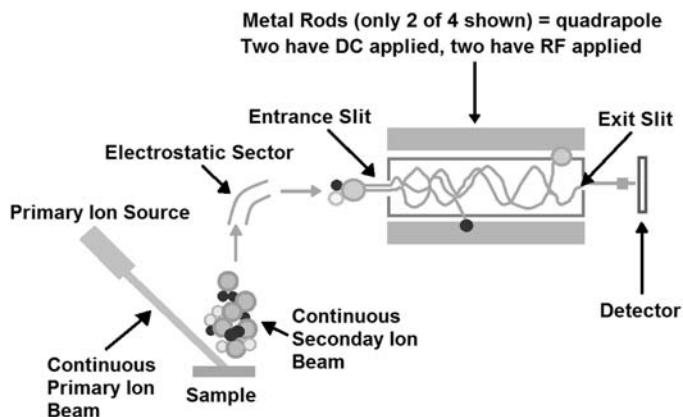


FIGURE 13.2-8 Schematic representation of a quadrupole-based SIMS instrument. *Used with permission from the authors.*

through the exit aperture and strike the detector. Quadrupole mass analyzers are not capable of the high mass resolution in which magnetic sector instruments are capable, but are quite robust and inexpensive.

13.2.5 Time-of-Flight-Secondary Ion Mass Spectrometry

Time-of-flight mass spectrometers [28] use a pulsed rather than a continuous primary ion beam (Fig. 13.2-9). Because of the pulsed nature of the beam, the total number of primary ions striking the surface during the course of an analysis is much lower than when using a magnetic sector or a quadrupole mass analyzer. This low primary ion dose makes TOF–SIMS ideally suited for the identification of organic contamination, because the higher doses of atomic primary ion species result in overlapping collision cascades that destroy the molecular character of the organic surface [35]. All secondary ions are accelerated to the same kinetic energy in the electrostatic acceleration field, but have different velocities, in the field-free drift region v proportional to the square root of their mass m :

$$\text{kinetic energy (KE)} = \frac{1}{2}mv^2 \quad (13.2-2)$$

By measuring the time t , hence “time-of-flight,” required for the secondary ions to travel over the fixed distance d from the sample surface to the detector, the mass/charge ratio can be determined from the measured velocity:

$$v = \frac{d}{t} \quad (13.2-3)$$

TOF–SIMS data may be presented in several ways: mass spectra are plots of detected intensity versus mass; depth profiles plotting the intensity of one or

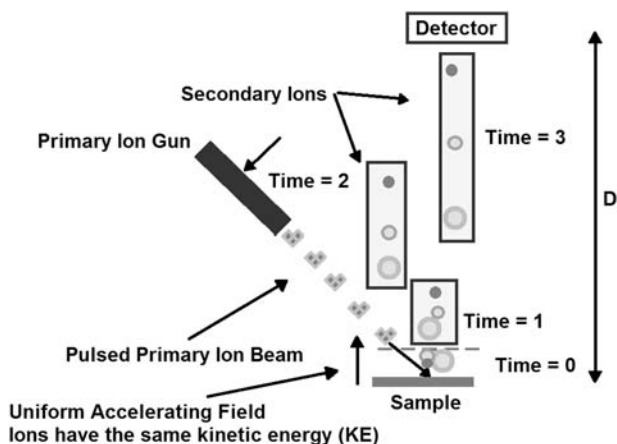


FIGURE 13.2-9 Schematic diagram of time-of-flight secondary ion mass spectrometry. Used with permission from the authors.

more ions as a function of depth into the sample; or as plots of an ion image, which show the intensity of a specific ion across the sample surface.

Because TOF—SIMS requires that the transit time of discrete secondary ion pulses be measured, the primary ion beam must also be pulsed. Additionally, unlike the mass filtering double-focusing or quadrupole instruments, in which all ions having a mass/charge ratio other than the desired m/z are filtered and discarded, the TOF employs parallel detection, in which all generated secondary ions are transmitted and eventually reach the detector. The lateral spatial resolution of TOF—SIMS is determined by the spot size of the ion beam used to bombard the sample surface. When performing a depth profile the distance the primary ion penetrates into the surface before stopping, the extent of the so-called “atomic mixing zone,” principally determines the depth resolution. With low-energy primary ions this distance can be kept as low as 1 nm. An advantage of TOF—SIMS is that two ion guns may be used for depth profiling; one for analysis and one for eroding the sample surface. The analysis ion gun can be optimized for spatial resolution while the sputter ion gun is optimized for depth resolution. In this way, 3-dimensional analysis can be done with both extremely high lateral and depth resolution.

TOF—SIMS offers several advantages over TXRF and VPD. Because of the decreased fluorescence yield for low atomic number elements, as shown in Table 13.2-2, conventional TXRF cannot readily analyze many species such as

TABLE 13.2-2 Detection Limits Achievable by TOF—SIMS of Various Elements on Si [40]

Element	TOF—SIMS Detection Limit (atoms/cm ²)
Li	1×10^7
B	2×10^7
Na	1×10^7
Mg	3×10^7
Al	2×10^7
K	1×10^7
Ca	1×10^8
Cr	1×10^8
Mn	5×10^8
Fe	5×10^8
Ni	1×10^9
Cu	1×10^9

Used with permission from *Surface Interface Analysis*.

Na, Li, Al, and F. Also, the detection limits of the transition elements measured by TXRF are typically in the low 10^{10} atoms/cm² range. Above this level, the two techniques give comparable results [36,37]. Additionally, the large analysis area (1-cm diameter) required for TXRF precludes small area analysis that is essential for equipment troubleshooting or device failure analysis.

VPD also holds the potential for higher sensitivity analysis; however, it has limitations as well. Contamination below the Si/SiO₂ interface may not be detected, because HF vapor does not dissolve the Si substrate. In addition, the collection efficiency of metals into the scanning droplet have varying efficiencies [7]. As with TXRF, application of VPD to non-Si surfaces and to small area analysis cannot be performed.

TOF–SIMS has the advantages of lower detection limits, capability of analyzing all elements, polyatomic, and organic compounds with very high spatial resolution. One difficulty is that isobaric (same mass) interferences exist in which several species reside at the same nominal mass with only small differences in their exact mass. However, recent generation TOF–SIMS instruments are capable of fully resolving major interferences without loss of sensitivity. Furthermore, TOF–SIMS has impressive surface contamination detection limits for those elements that are of particular importance to semiconductor processing. Douglas and Chen [38] have summarized the detection limits for a wide range of analytes achievable using state-of-the-art TOF–SIMS instrumentation, which are shown in Table 13.2-2. An advantage of current TOF–SIMS technology is that the routine analyses of Si wafers has been automated to the point where multiple samples can be analyzed with sample positioning, data collection, analysis, and report preparation being performed in a fully unattended mode. In addition to quantitative elemental contamination, work by Chartogne and Saldi [39] has shown promise for determining the chemical state (e.g., whether Fe is present as FeO or Fe₂O₃) based on a measurement of the relative ratios of various molecular fragment ions (FeO⁺, FeO₂⁺, and FeO₃⁺).

The accuracy of the analysis is determined by the accuracy of the standards used to calibrate the SIMS instrument. The development of SIMS standards is currently an area of active research [40]. A comparative study of Na contamination on bare Si wafers using surface contamination standards was performed using TOF–SIMS versus double-focusing dynamic SIMS analysis of the same wafers for ion implanted reference samples. The results are shown in Fig. 13.2-10; excellent correlation is observed over several orders of magnitude.

An interesting application of TOF–SIMS imaging analysis to circumvent the relatively small raster area of current instrumentation, relative to that of an entire wafer, involves analysis of a large (mm × mm) area of a wafer surface by creating a montage of a large number of small area images obtained by mechanically moving the sample stage during analysis [41]. An example of

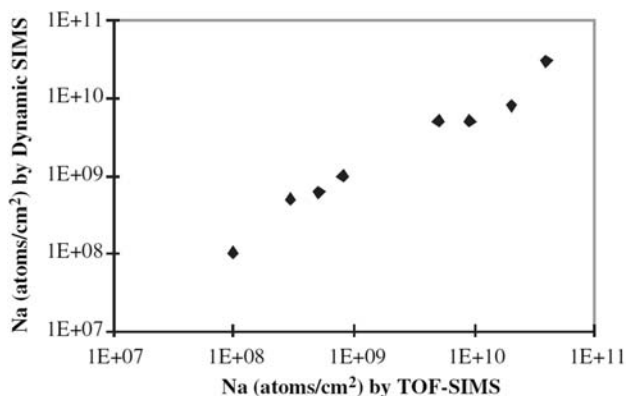


FIGURE 13.2-10 Comparison of surface Na surface contamination levels as determined by TOF-SIMS and dynamic SIMS [34]. *Used with permission from Future Fab International.*

this method is provided in Fig. 13.2-11, which shows the surface of a wafer intentionally contaminated with polydimethylsiloxane. The small image shows an analysis of the characteristic SiC_3H_9^+ ion fragment obtained at the largest raster area of the instrument ($500\ \mu\text{m} \times 500\ \mu\text{m}$). The large area image is then generated from 576 separate rastered areas, each obtained by a precise mechanical translation of the full wafer.

One approach to increase the sensitivity of TOF-SIMS is to increase the flux of primary ions to the surface of the sample. This approach is useful until the intensity of the matrix species used for quantification exceeds the linear range of the detector resulting in count losses and the consequent errors in

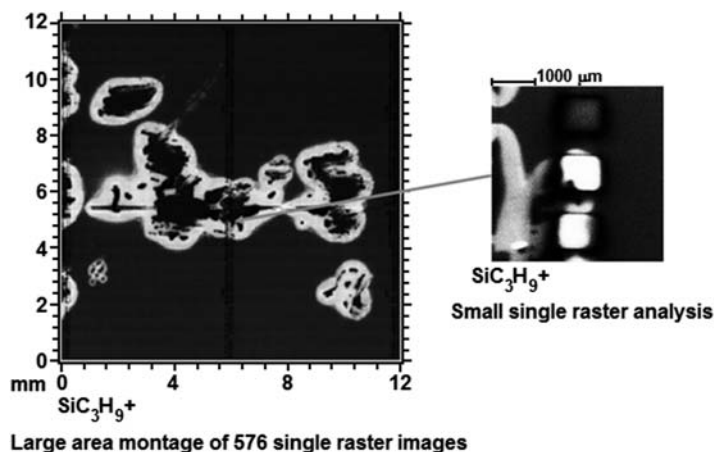


FIGURE 13.2-11 TOF-SIMS image of the polydimethylsiloxane fragment ion SiC_3H_9^+ illustrating the capability of large area analysis. *Used with permission from the authors.*

quantification. A novel solution to this quandary is the extended dynamic range [42] approach in which the secondary ion pulse from the matrix species is diverted into an alternate path in the flight tube from the analyte ion pulse. This alternate path includes a physical plate containing a series of apertures that allow only a fraction of the matrix ions to pass onto the detector. This reduces the matrix ion intensity by a known fraction determined by the aperture/total plate surface area ratio to the linear response range of the detector. The matrix total ion intensity is then accurately measured and used to calculate the concentration of the analyte.

13.2.6 Energy Dispersive X-Ray Spectroscopy

EDS [43] is an X-ray fluorescence technique in which the excitation source used to generate the core electron vacancies, as shown in Fig. 13.2-2, is a beam of energetic electrons, commonly from a scanning electron microscope (SEM). The energies of the resulting X-ray fluorescence photons is measured in the same manner as in TXRF, typically with an SiLi detector, with detection limits on the order of 0.1–1 at.%. EDS is a fast, nondestructive analytical technique that is relatively inexpensive and readily adapted to most SEM instruments. EDS is quite useful for the analysis of particulate and other localized forms of contamination. Spectra of the contamination distribution are obtained by monitoring the detected X-ray intensity at the energy characteristic of a particular element as a function of the scanned electron beam position. However, it must be noted that the high-energy electrons generate X-ray at a considerable depth into the sample [44] (up to 3–5 μm) and the fluorescent X-rays have an escape depth well beyond this thickness. This is particularly apparent when a normal (perpendicular to the surface) electron beam is used. Therefore, when analyzing small particles, $\sim 0.2\ \mu\text{m}$ in diameter or less on a SiO_2 film, the majority of the analytical signal is generated from the underlying film and the particle may be misidentified as a silica particle.

13.2.7 Auger Electron Spectroscopy

AES (for an excellent review of AES, please see Ref. [45]) is a technique in which a primary energetic electron beam is directed onto a sample and interacts with atoms on the sample surface by causing the ejection of a core photoelectron, as shown in Fig. 13.2-2. Similar to TXRF, an outer shell electron fills the vacancy created by the ejected core electron, leaving the atom in an excited energy state. For lower atomic number atoms, this energy is relieved by the ejection of an Auger electron, whose kinetic energy is estimated from the differences in binding energies of their energy levels. Auger transitions are commonly “named” by the electron levels that are involved in the transition. An example is a KLL transition involving an initial K level hole being created, with an L shell electron filling the K shell hole, and another L

shell electron being ejected as the Auger electron. Also similar to TXRF, the energy of the Auger electron reveals the identity of the atom and the detected intensity provides concentration information, as shown in Fig. 13.2-12. Because of the low intensities of the Auger peak maxima, the data are typically plotted as the first derivative of the intensity versus energy, to enhance the detectability of the signal.

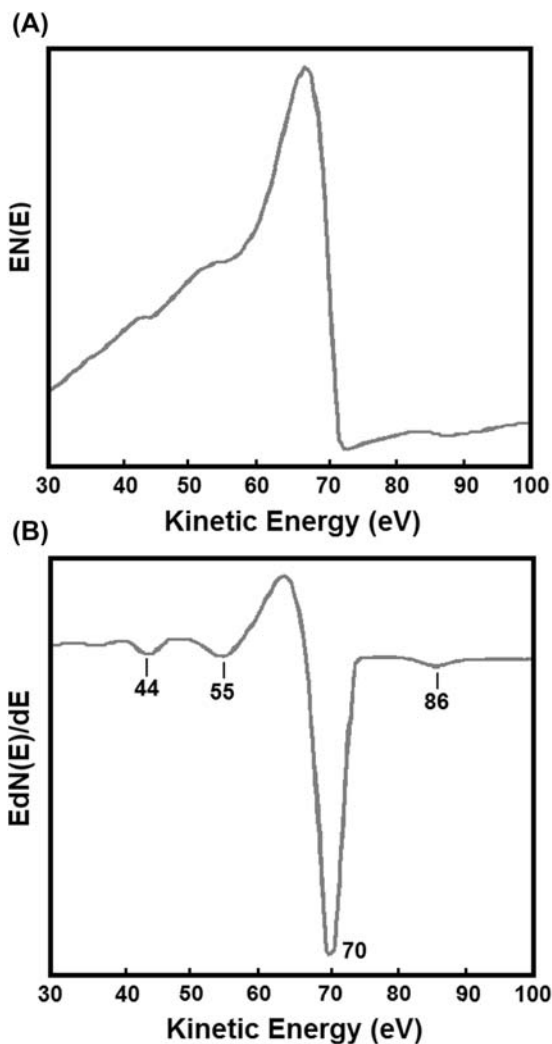


FIGURE 13.2-12 The 70-eV LMM Auger transition signature of Al shown as (A) intensity and (B) as the first derivative of intensity versus kinetic energy [17]. *Used with permission from Handbook of Auger Electron Spectroscopy, Third Edition.*

One of the main strengths of AES is that the primary electron beam is focused to a very small diameters and is well suited to the analysis of particulate contamination on Si wafers. As with TOF—SIMS, AES may be used in a spectral survey mode to qualitatively determine the elements present in the area of interest. AES also may be used in a depth profile mode integrated with an ion sputtering beam to determine the depth of elemental contaminants, or in an elemental mapping mode to produce images in which the contrast is related to the lateral distribution of the element.

13.2.8 X-Ray Photoelectron Spectroscopy

XPS (for a comprehensive review of XPS, see Ref. [46]), also known as electron spectroscopy for chemical analysis, is an elemental analysis technique in which core electron vacancies are created by primary X-ray radiation and the kinetic energy of the ejected core photoelectron is measured. Each element has a characteristic binding energy associated with its core electron levels. The measured kinetic energy, of the ejected photoelectron is determined by the incoming energy of the incident X-ray photon, $h\nu$ (typically Mg or Al $K\alpha$ radiation are used). The binding energy, BE, of the photoelectron is calculated by the relationship:

$$KE = h\nu - BE \quad (13.2-4)$$

BE is dependent on the chemical environment of the atom, e.g., carbon in C_2H_6 has a different binding energy than carbon in CH_2F_2 . This effect, commonly termed the chemical shift, gives rise to the greatest forte of XPS, the ability to provide information about the chemical bonding of the analyzed element. In order to measure these chemical shifts, the use of a high-energy resolution hemispherical analyzer is required for XPS. Detection limits are similar to those of AES (0.1–1 at.%). XPS may also be used in survey, depth profiling (when used with sequential ion sputtering), and imaging modes. However, because of the difficulties inherent in focusing a primary X-ray beam, the lateral resolution of XPS is considerably less than that of AES.

13.3 ORGANIC SURFACE CONTAMINATION

13.3.1 Introduction to Organic Contamination Analysis

The analysis of organic (carbon-containing) contamination presents a number of challenges, the first being that it is not only the identification of the elements present, but also their specific arrangement to each other. Organic molecules contain predominately C, O, N, and H, but may contain other elements, including Si. However, molecules with different atomic arrangements of this limited variety of atoms may have profoundly different properties.

13.3.2 Time-of-Flight-Secondary Ion Mass Spectrometry

Providing information of the arrangement of elements within an organic molecule is one strength of TOF–SIMS. When a primary ion impacts the surface, molecules within the area immediately surrounding the impact site are shattered into their composite atoms. A little further away from the initial ion impact site, the collision cascade has lost sufficient energy that the molecules undergo slightly less energetic collisions and small fragments of several atoms survive. Much farther away, larger fragments survive, until at some distance the intact molecule, referred to as the parent molecule is ejected off the surface with interactions that are insufficient to disrupt atomic bonds within the molecule and the resulting ion mass is reflective of the original intact molecule. The mass of each of the fragments and their relative intensities are determined by the arrangement of the atoms in the molecule. By analogy to a metal chain, the strength of the bonds determines the distribution of weak links and the size of each fragment of the chain when the chemical bond fails. Comparing the mass and the relative intensity of each fragment with that of known substances, the identification of an unknown material can be determined. Even if a reference mass spectrum of the unknown material is not available the information about its nature based upon the knowledge of a similar fragment may be used to deduce the compound.

Upon inspection of the spectral analysis of the unknown particle in Fig. 13.3-13, it could be deduced that the main intensity maxima of the

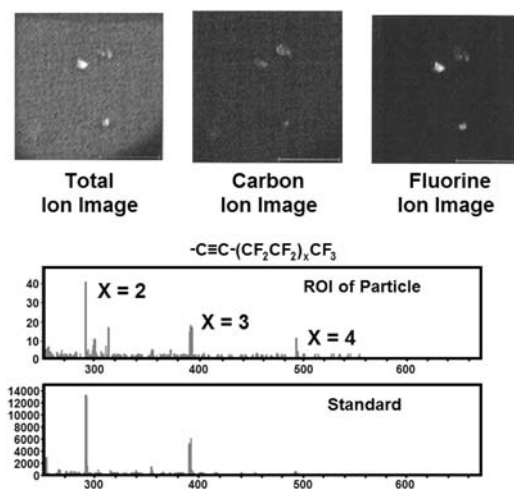


FIGURE 13.3-13 Analysis of particle contamination using TOF–SIMS. Ion images of the particles are shown in the upper portion. A comparison of the mass spectrum of the particles obtained with a region-of-interest (ROI) analysis using a reference spectrum obtained from a suspected source of the particle is shown in the lower portion [34]. *Used with permission from Future Fab International.*

material indicates a fluorocarbon. TOF—SIMS operating in the static mode acquires a mass spectrum of the sample surface where each ion impacts; the resulting mass spectrum is reflective of the undamaged surface. In dynamic SIMS, the static ion flux limit is quickly exceeded and the mass spectrum begins to show the sputtered damage surface, which can be different compared to the undamaged surface (e.g., all polymers begin to show a mass spectrum reflective of a graphite surface).

The impact of organic molecular contamination on process yield and device reliability is a major concern for advanced semiconductor device manufacturing. Of concern is not only the level of these contaminants in the manufacturing environment (commonly termed atmospheric molecular contamination), but more importantly the fraction that adheres to the wafer surface. TOF—SIMS is one of the techniques, along with thermal desorption, that allows analysis of low-level organic contamination on wafer surfaces. TOF—SIMS has the unique advantage that it also provides high lateral resolution, permitting the determination of quantities on different areas of patterned wafers. This is especially useful in the analysis of organic particle contamination. An example is shown in Fig. 13.3-13. The image shown is the sum of all secondary ions generated at each pixel and provides topography information similar to that seen in a secondary electron micrograph. The C and F images reflect the intensity of these ions and reveal the particles are rich with these two elements, relative to the surrounding area.

A new area revolutionizing TOF—SIMS organic analysis is the development of cluster primary ion sources, starting with the use of small cluster ions, such as C_{60}^+ ions [47], and evolving into massive cluster ions, such as Ar_{400}^+ [48]. These cluster ions deposit energy in a means that sputters the surface, but does not destroy the molecular character of the surface allowing for the depth profiling of organic samples [49]. Additionally, these cluster ions significantly enhance the ion yield of larger molecular fragments with enhancements of two to four orders of magnitude for C_{60}^+ relative to Ga^+ primary ions [50]. Furthermore, a decrease in fragmentation of large organic molecules [51] allows a more detailed analysis of the structure of large molecules on organic surfaces.

A unique strength of TOF—SIMS is the ability to analyze polyatomic (e.g., $Fe_2O_5^+$) and organic compounds simultaneously with elemental analysis. Organic materials are critical contaminants, particularly in areas such as gate oxide performance as described in a Sematech report 95052812A-TR [46]. TOF—SIMS has the ability to achieve high mass resolving powers without a major loss in instrument transmission (the ratio of total secondary ions generated at the sample to those that impact to the detector), compared to other mass spectrometric techniques. Mass resolving powers ($M/\Delta M$) of 10,000–15,000 are readily achievable, providing the ability to easily separate isobars of the same nominal mass; an example of mass 30 amu region for an Si surface is shown in Fig. 13.3-14.

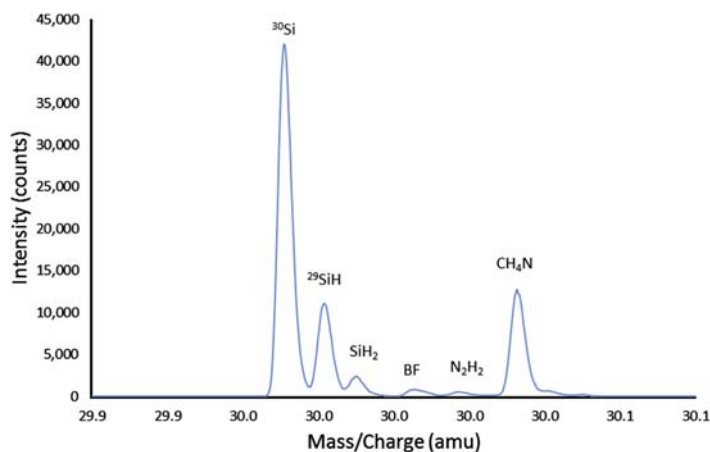


FIGURE 13.3-14 Separation of isobars at a nominal mass of 30 amu by TOF–SIMS. *Used with permission from the authors.*

One issue concerning TOF–SIMS, as with any analytical technique performed in vacuum, is volatile contaminants may be desorbed from the wafer surface during the evacuation of the chamber, and thus may not be detected during analysis. This problem may be addressed by the use of a cold stage, in which the sample is brought to near liquid N_2 temperature prior to evacuation, so that its vapor pressure is significantly reduced, limiting the loss from the surface.

Another question that is involved in the application of TOF–SIMS is the absolute (in terms of molecules/cm²) rather than relative (i.e., sample A has more than the control sample B) quantification of organic contamination levels. As with all types of SIMS analysis, this requires the use of standard reference samples of known concentration on the substrate of interest. These types of standards are difficult to prepare to obtain a homogenous distribution and known concentration. One approach is to use inkjet printer technology to deposit uniform droplets of a standard solution over an area large enough to be measured with TOF–SIMS [52].

13.3.3 Fourier Transform Infrared Spectroscopy

FTIR is a method in which light from the infrared (IR) portion of the electromagnetic spectrum interacts with molecules on the surface of a wafer. The position of the atoms composing the molecules are not static but can move relative to other atoms similar to the way billiard balls might move if fastened to each other by springs. The movement of the atoms in molecular bonds occurs at specific frequencies. When IR radiation at the vibration frequency of the chemical bond is directed at the molecule, the bond absorbs the light at

these frequencies and thus does not pass into the IR detector. By scanning the IR frequencies and plotting the intensity (the transmitted light), or its inverse (the absorbed light), an IR spectrum is generated, as shown in Fig. 13.3-15.

From the frequencies of the absorption maximum, the type of chemical bonds present in the molecule may be deduced. The pattern of peaks in the spectrum is a characteristic of a particular molecule. From the intensity of the characteristic peaks, the concentration can be determined by referencing the unknown molecule signal to a standard sample signal of known concentration. For the analysis of contaminant films on wafer surfaces, a special type of FTIR analysis is commonly performed [53]—ATR—FTIR analysis. ATR—FTIR directs an IR beam onto a crystal sample holder, so that the beam strikes the crystal surface below the critical angle and undergoes total reflection, analogous to TXRF. The beam is totally reflected off the crystal surface and creates a standing wave that penetrates beyond the crystal surface on the order of 100 Å. When a sample, such as a section of a Si wafer, is placed in contact with the crystal surface the standing wave is absorbed by molecules on the wafer surface and the absorption is detected as a reduced intensity at the detector for those wavelengths absorbed by the species on the wafer surface.

The FTIR may also be employed in microscopy mode, in which the absorption of IR radiation is localized to a specific region of the sample, such as the area occupied by a particle or other contaminant. FTIR images may also be obtained in which the contrast of the image is modulated by the absorption of one or more frequencies and provide an image of the distribution of a particular species across the sample. FTIR may be applied to study adsorbed organic contaminant films on Si surfaces [54]. When using an ATR—FTIR instrument with a high number of internal reflection passes (~ 300), the detection limit in the order of 10^{12} C atoms/cm² has been reported [55].

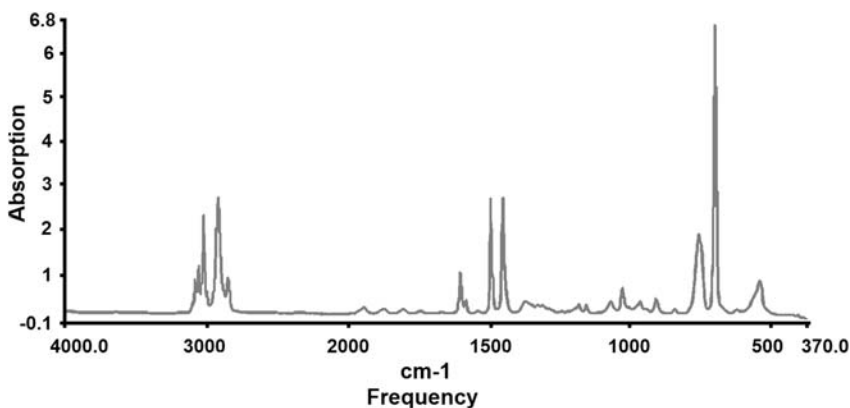


FIGURE 13.3-15 FTIR absorption spectrum of polystyrene. *Used with permission from the authors.*

However, while it is not as sensitive as other surface techniques, such as TOF–SIMS, when employed in a microscopy mode, is particularly useful for large-size ($>10\text{ }\mu\text{m}$) particulate contamination. FTIR provides information on chemical bonding and is useful as a front-line qualitative technique to identify unknown contaminants [56]. FTIR may also be used to measure characteristics of inorganic films on Si wafers, such as ultrathin SiO_2 layers [57].

Another IR technique is IR reflection absorption spectroscopy (an overview of this technique is provided in reference [58]), which is used to identify the functional groups, as well as their orientation on the surface. In this technique the incident IR radiation is reflected from a highly reflective (typically a metal) surface at grazing incidence angles. Unfortunately, this technique is more suited for highly reflective metal surfaces, rather than semiconductor surfaces on which it demonstrates reduced sensitivity. However, methods that can overcome this limitation by using a semiconductor layer deposited on a highly reflective metal surface have proven to be useful [59].

13.3.4 Thermal Desorption Gas Chromatography Mass Spectrometry

An alternate means for the analysis of organic species on surfaces involves heating a wafer in a sealed chamber and collecting the resulting gas phase contaminants in a tube containing an adsorbent, such as activated C or Tenax [60]. The tube is rapidly heated and the desorbed contaminants are introduced onto a gas chromatograph column where the contaminants are separated based upon their relative affinities to the stationary phase coating inside the chromatographic column. Those having a higher affinity are retained on the stationary phase at a higher degree and, consequently, spend a shorter period of time in the mobile gas phase moving towards the end of the column. Contaminants exit, or elute, from the column at different times and are then analyzed using MS, where they are fragmented, typically by the impact of energetic electrons, and then mass analyzed using a quadrupole mass analyzer. The resulting mass spectrum is analogous to that produced by TOF–SIMS. Detection limits on the order of 5×10^{11} C atoms/cm² have been reported [60]. This method has advantages over TOF–SIMS because the entire wafer surface is sampled and any potential loss of contaminants during sample introduction vacuum is eliminated. It suffers from the disadvantage that lateral distribution of the contaminant is lost.

Thermal desorption spectrometry may also be performed by incorporating a small quadrupole mass analyzer, typically a residual gas analyzer, into the vacuum chamber used to heat the wafer sample to monitor the evolved gases as a function of temperature, as shown in Fig. 13.3-16. This may even be performed in a production tool that normally operates at atmospheric pressure by the incorporation of an atmospheric pressure ionization mass spectrometer [61]. However, unlike TOF–SIMS, with the relatively low mass resolution of

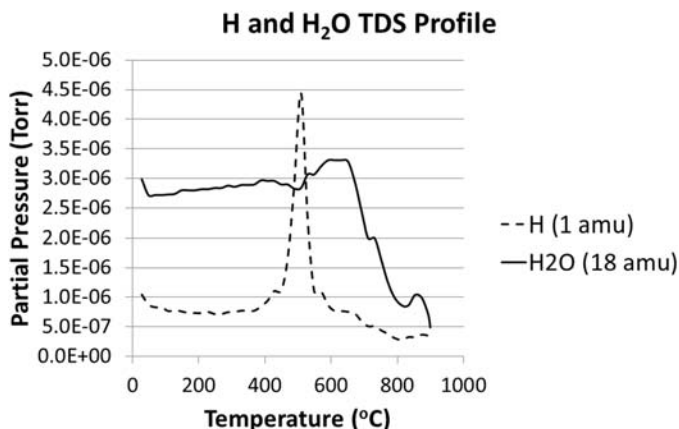


FIGURE 13.3-16 TDS temperature profile obtained with a quadrupole mass analyzer directly in the vacuum chamber. *Used with permission from the authors.*

the quadrupole mass filter not allowing the separation of molecular fragments nor having the separation step of the gas chromatograph, unambiguous identification of desorbed molecular species, especially at higher masses, is difficult. The advantage of this method, however, is that the temperature dependence of the evolved species may be obtained, rather than the integration overall temperatures of the adsorbent tube method. Although multiple adsorbent tubes may be used to measure volatile species evolved over a wide temperature range, performing this analysis can be complex and will not have the temperature resolution of the direct measurement technique.

13.4 SURFACE MORPHOLOGY ANALYSIS TECHNIQUES

13.4.1 Atomic Force Microscopy

A critical concern regarding Si wafer surfaces, apart from chemical contamination, is roughness. With gate dimensions approaching 20 Å, measurement of roughness on a nanometer scale is imperative for device yield and performance. This roughness may be preexisting on incoming wafers or develop as a result of processing, such as a post-etch wet cleaning step. Chapter 4 thoroughly outlines the surface conditioning requirements, while Chapter 9 outlined the processes that roughen the surface. Atomic force microscopy (AFM) [62] is a technique that is capable of measuring the 3-dimensional surface roughness with angstrom resolution. AFM operates by mechanically scanning a sharpened probe tip attached to the end of a spring cantilever, using piezoelectric actuators. As the tip scans laterally over the wafer surface it interacts with the attractive (noncontact mode) or repulsive (contact mode) forces between the tip and surface and moves up and down following the

contours of the surface. The size of the AFM tip has a significant impact on the measured roughness [63]. The up and down displacement of the tip is measured by a laser beam reflected off the tip onto a quadrant photodetector array. A 3-dimensional image (Fig. 13.4-17) is then constructed from the x - y position of the tip (determined from the known voltages applied to the positional actuators) and the vertical displacement of the tip (determined from the output of the photodiode array). The x - y displacement is displayed in 2-dimension and the height is shown as color contrast in the image.

Roughness is typically presented by a mathematical relationship known as root-mean-square. This value describes the variation of the surface height around the average surface height. A limitation of AFM is that the analyzed area is quite small with respect to the entire wafer surface. Unless the wafer roughness is known to be highly homogenous, analysis of a small area of the wafer may not be representative of the entire wafer surface. However, this small analysis area is an advantage when a measurement of the roughness across a small feature is desired with high lateral resolution. For generalized roughness measurement across the entire wafer surface, metrologies such as light scattering measurements, in which the roughness is inferred from the degree of light scattered from the surface, are also available.

13.5 FUTURE CONTAMINATION ANALYSIS TECHNIQUES AND TRENDS

The advancement of IC manufacturing technology requires a commensurate advancement in the sensitivity of contamination analysis and hence this leads

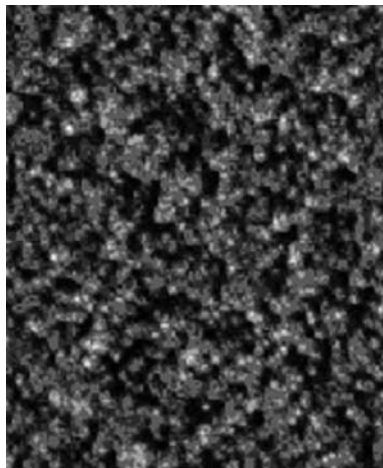


FIGURE 13.4-17 AFM measured roughness of a wafer surface; total height range shown is 3 nm. *Used with permission from the authors.*

to new and optimized analytical methodologies. New metrology development must be considered from two perspectives: (1) effective measurement of the total contamination on the surface a wafer and (2) the need to differentiate the background contamination from the sample contamination. To effectively measure the contamination across the entire wafer surface requires the use of preconcentration methods such as VPD that assist the detection limit requirements of the analytical metrology, as the contamination is collected into a concentrated, smaller volume. This, by its nature, results in loss of lateral distribution information, which may be useful in determining the source of the contamination. For other analytical techniques that do not allow preconcentration of the contaminant, the impurity level must be great enough for detection. If analysis must be performed on a wafer with device features, the analysis may be volume-limited, for example if the analysis volume is 100 nm^3 and the contamination is present at a level less than 1 at.%, then the number of atoms available for analysis is below detection limits. This is particularly true for destructive metrologies, such as SIMS, as opposed to methods such as AES, where the same analytical volume may be sampled many times. Additionally, the purity of chemicals, vessels, and the laboratory ambient environment must be concurrently improved to provide low levels of background contamination.

Because of the impact of airborne basic species on acid-catalyzed photoresist, condensed dopant atoms on the active regions of devices with ever shrinking physical dimensions, and the impact of condensed molecules that may form compounds, such as nitrides and oxides, the gate layer requires an increased emphasis on the analysis of organic contamination in advanced semiconductor device manufacturing [64,65]. The sensitivity of mass spectrometric techniques, combined with their superb specificity for qualitative analysis for organics, makes them well suited for this application. Combined with sample preconcentration similar to VPD, TOF–SIMS with a sample cold stage should make an excellent technique, especially when combined with sensitivity enhancement with the use of novel cluster primary ions [66].

13.6 SUMMARY

Measurement of ultratrace contamination is critical for optimal revenue generation in semiconductor fabrication. Routine measurement of baseline levels must be performed so that contamination trends are noted and addressed prior to yield impact. Qualitative determination may provide valuable insight regarding contamination sources and mechanisms. A wide variety of metrology techniques is available to measure wafer surface contamination, each having unique strengths and weaknesses. Consideration of the information required for contamination remediation will determine the most appropriate technique. With decreasing device dimensions, the need for both higher spatial and detection sensitivity will certainly follow the same trend.

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